4.4 Indeterminate Forms and L'Hôpital's Rule

Name _____ Period ____ Date _____

AP Calculus 4.4 (30-minute plan) · Student sheet

I can…

- identify an indeterminate form.
- use L'Hôpital's Rule for the forms 0/0 and ∞/∞.
- rewrite other indeterminate forms so L'Hôpital's Rule applies.

Vocabulary · fill in the missing word

Indeterminate form — a limit like $\frac{0}{0}$ or ∞/∞ whose value can't be found by _____ alone

L'Hôpital's Rule — if the limit of $\frac{f}{g}$ is $\frac{0}{0}$ or ∞/∞, it _____ the limit of $\frac{f'}{g'}$

Other indeterminate forms — _____

Part 1 · Notes and examples (fill in each note, then work the example)

L'Hôpital's Rule: if the limit of f(x)/g(x) is 0/0 or ∞/∞, then it equals the limit of _____.

Example 1

Find the limit as $x \to 0$ of $\frac{\sin(3x)}{2x}$.

Check the form first: use the rule only for 0/0 or _____.

Example 2

Find the limit as $x \to 2$ of $\frac{x^3 - 8}{x - 2}$.

Use it again: if the new limit is still 0/0, apply L'Hôpital's Rule _____.

Forms 1^{∞}, 0^0, ∞^0: take ln y, find its limit L, then the answer is _____.

Example 3

Find the limit as $x \to \infty$ of $\frac{x^2}{e^x}$.

Part 2 · Partner practice

Partner practice

(AP® Practice) The limit as $x \to 0$ of $\dfrac{e^x - 1}{\sin(2x)}$ is

(A) $\dfrac{1}{2}$ (B) 2 (C) 0 (D) does not exist